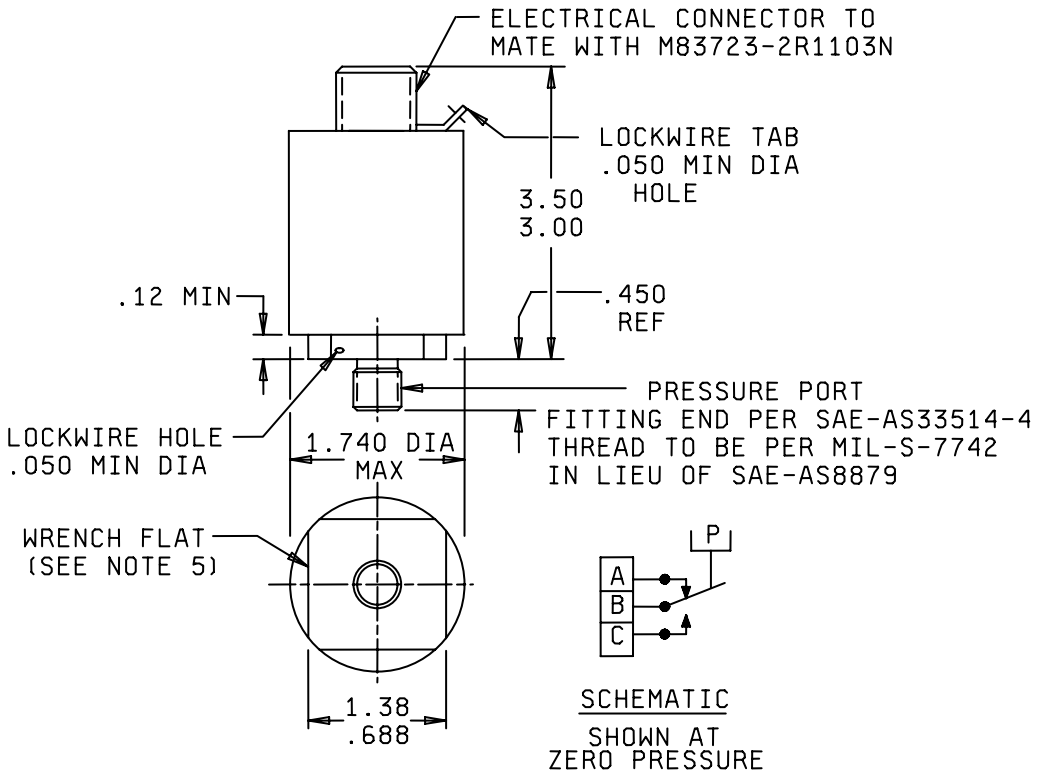
INCH-POUND
MIL-DTL-9395/30F
14 December 2004
SUPERSEDING
MIL-DTL-9395/30E
30 May 2001

DETAIL SPECIFICATION SHEET

SWITCHES, PRESSURE, (GAGE), TYPE II, 5 AMPERES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-DTL-9395.

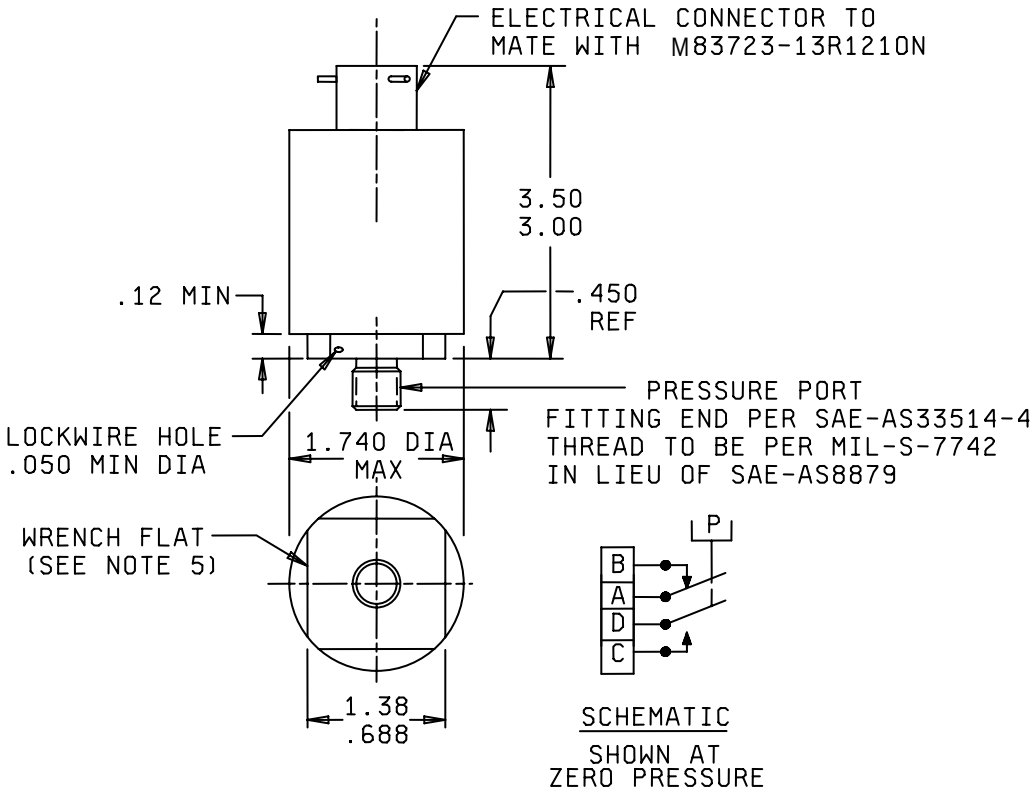


WEIGHT: NOT TO EXCEED .45 LBS.

Configuration 1

FIGURE 1. Switches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:16Z
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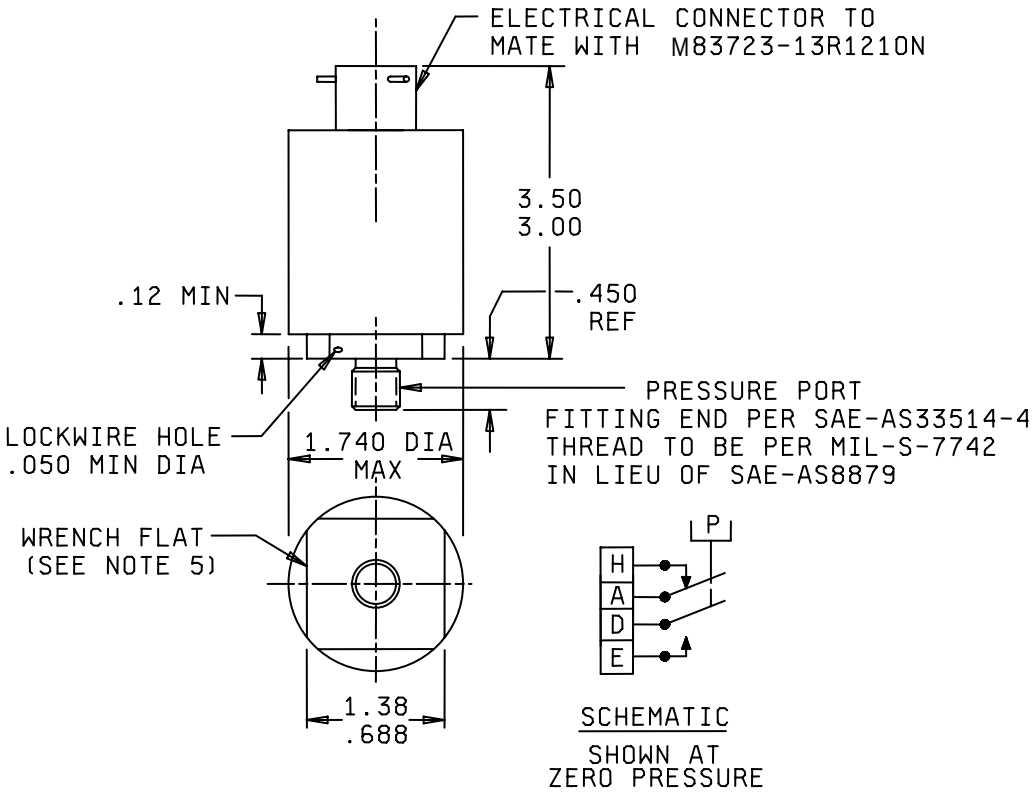


WEIGHT: NOT TO EXCEED .45 lbs.

Configuration 2

FIGURE 1. Switches - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:16Z
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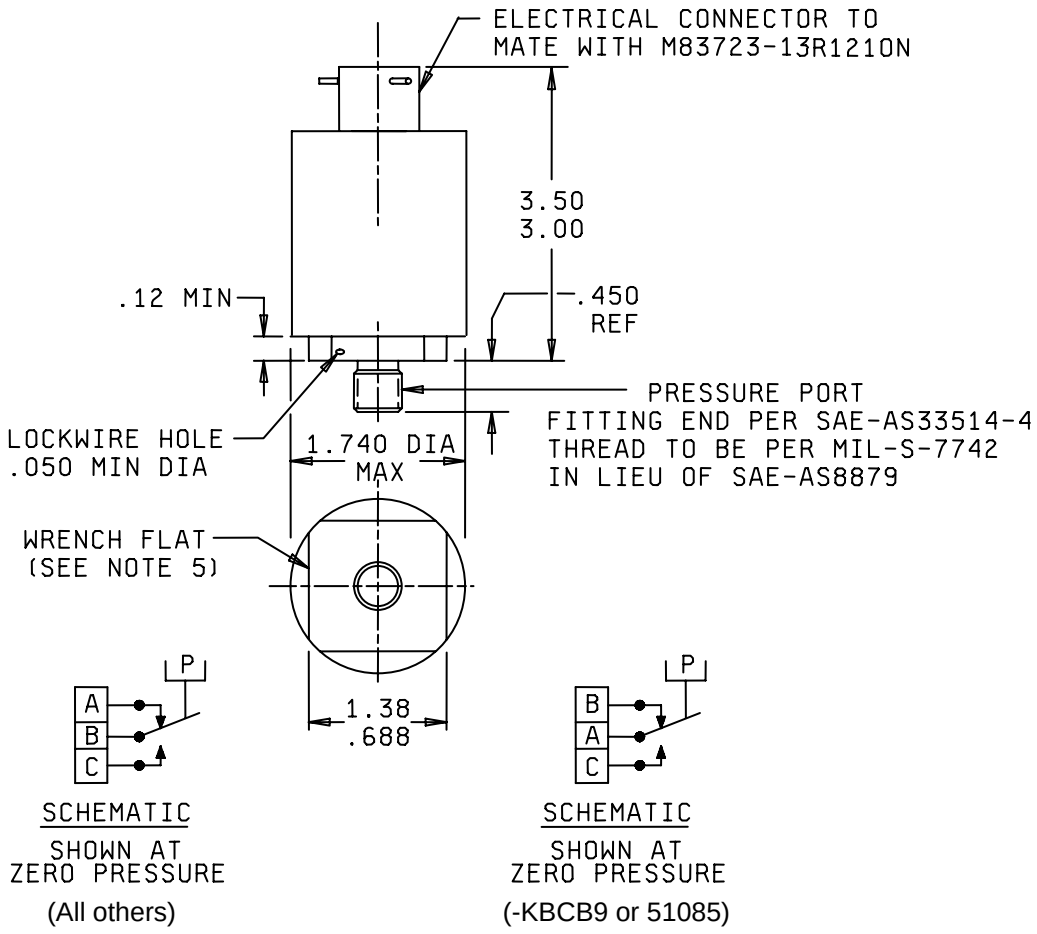


WEIGHT: NOT TO EXCEED .45 lbs.

Configuration 3

FIGURE 1. Switches - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:16Z
Check the source to verify that this is the current version before use.

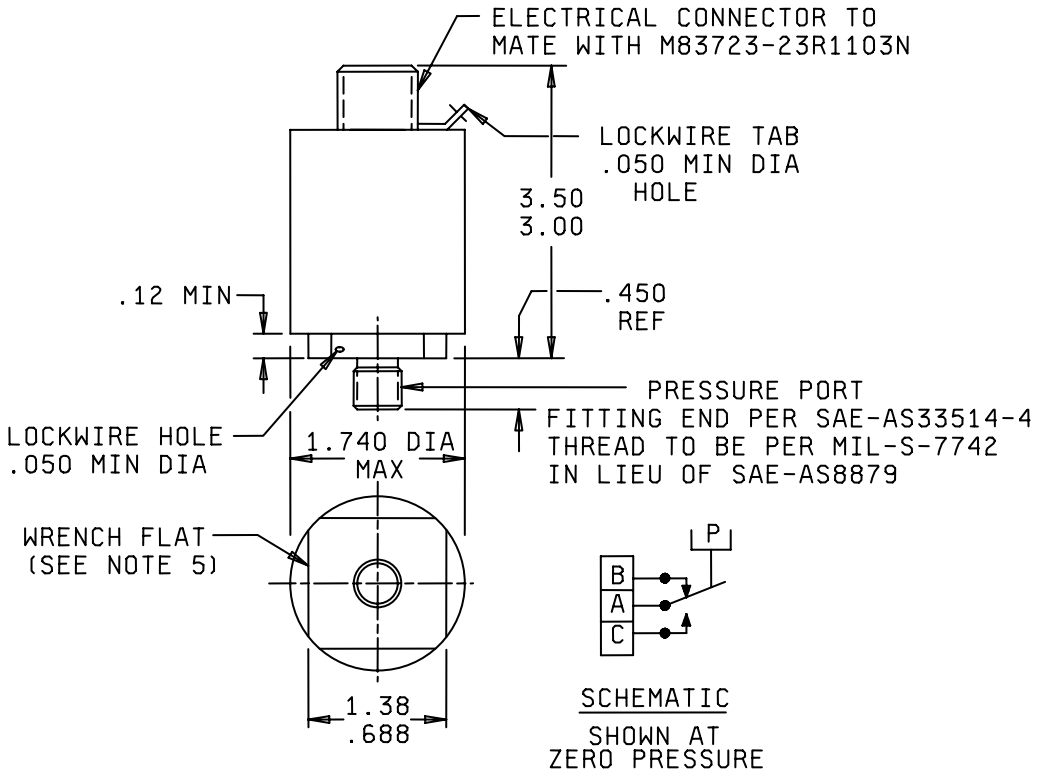


WEIGHT: NOT TO EXCEED .45 lbs.

Configuration 4

FIGURE 1. Switches - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:16Z
Check the source to verify that this is the current version before use.

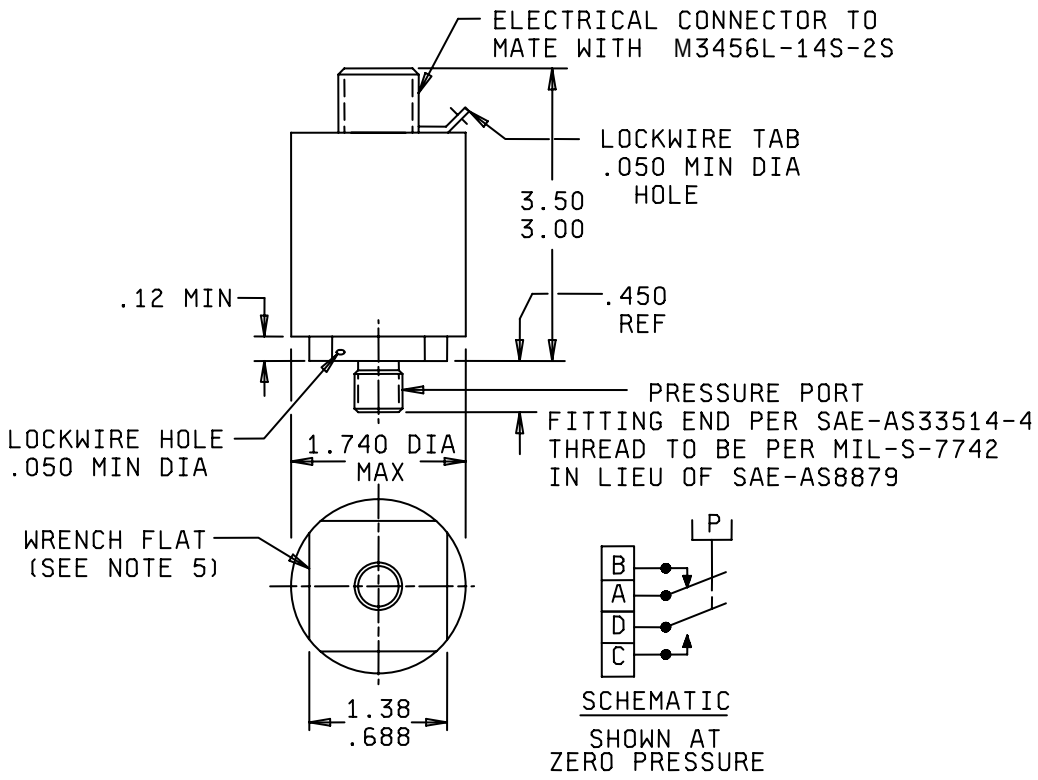


WEIGHT: NOT TO EXCEED .45 lbs.

Configuration 5

FIGURE 1. Switches - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:16Z
Check the source to verify that this is the current version before use.

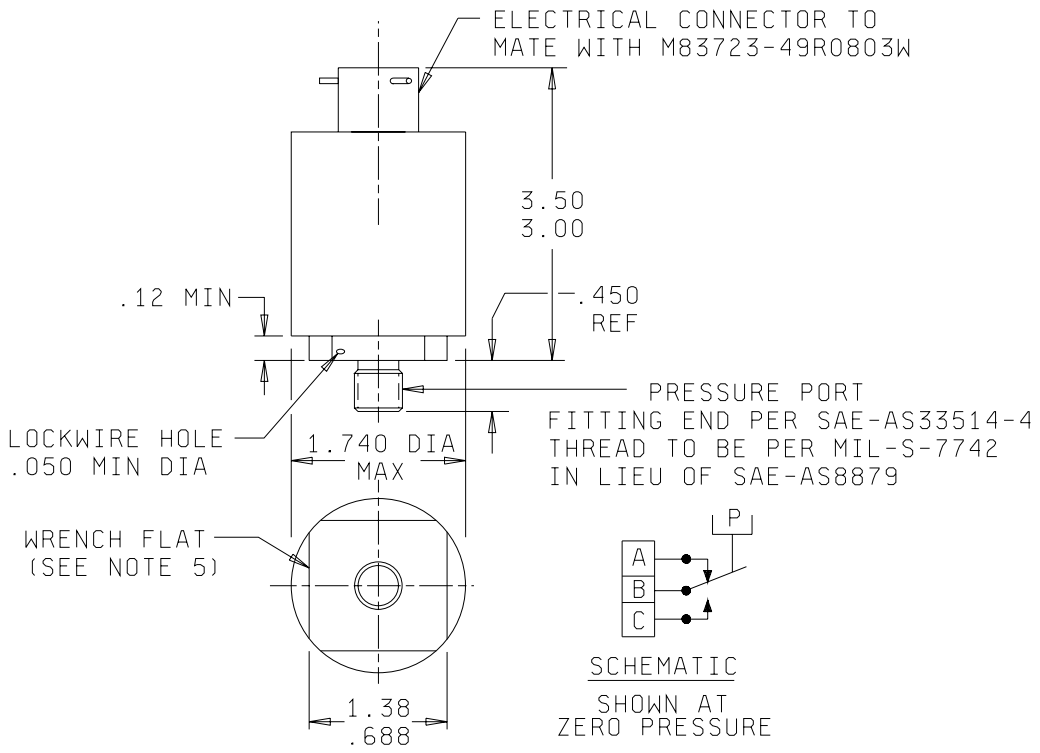


WEIGHT: NOT TO EXCEED .45 lbs.

Configuration 6

FIGURE 1. Switches - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:16Z
Check the source to verify that this is the current version before use.

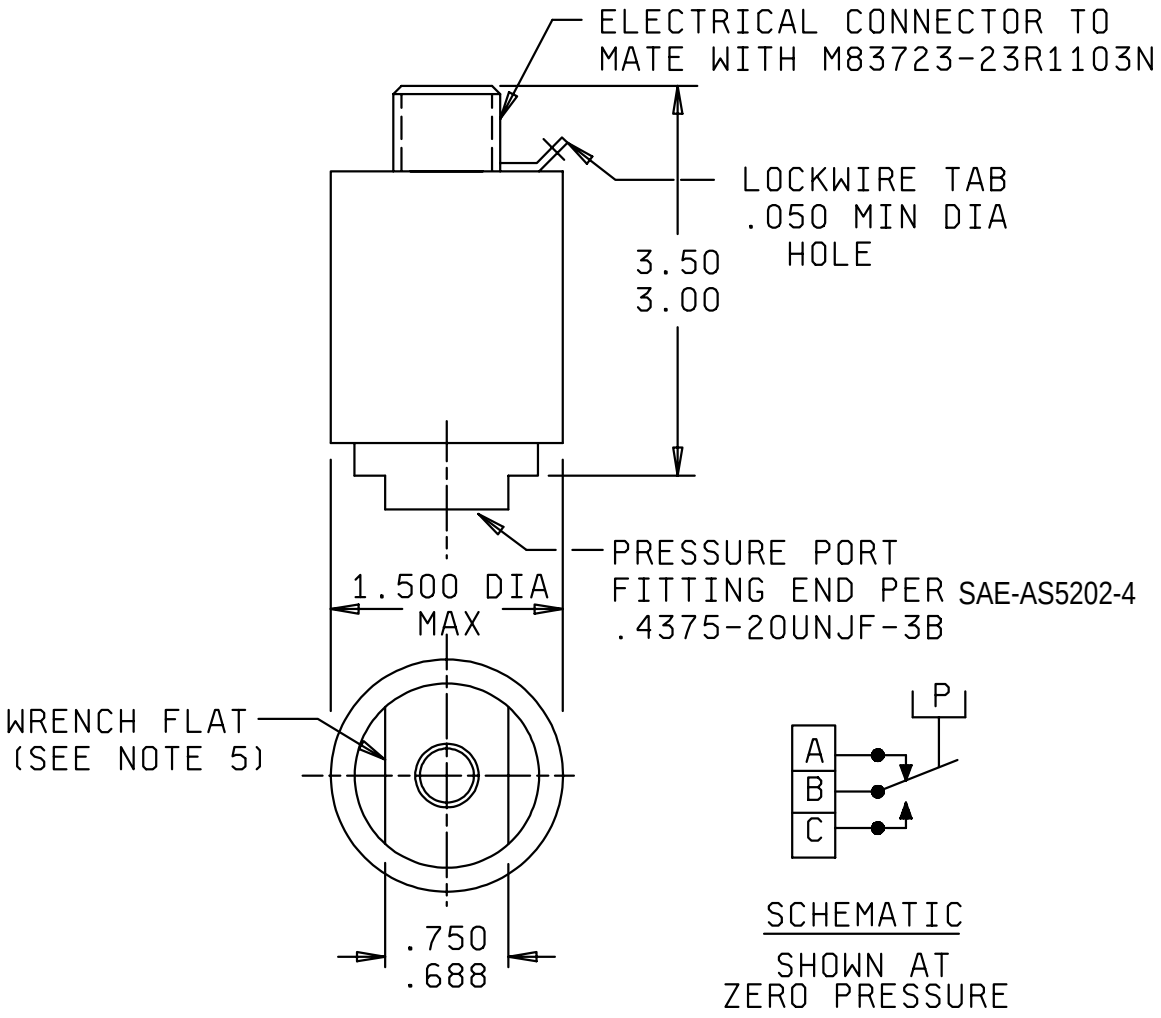


WEIGHT: NOT TO EXCEED .45 lbs.

Configuration 7

FIGURE 1. Switches – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:16Z
Check the source to verify that this is the current version before use.

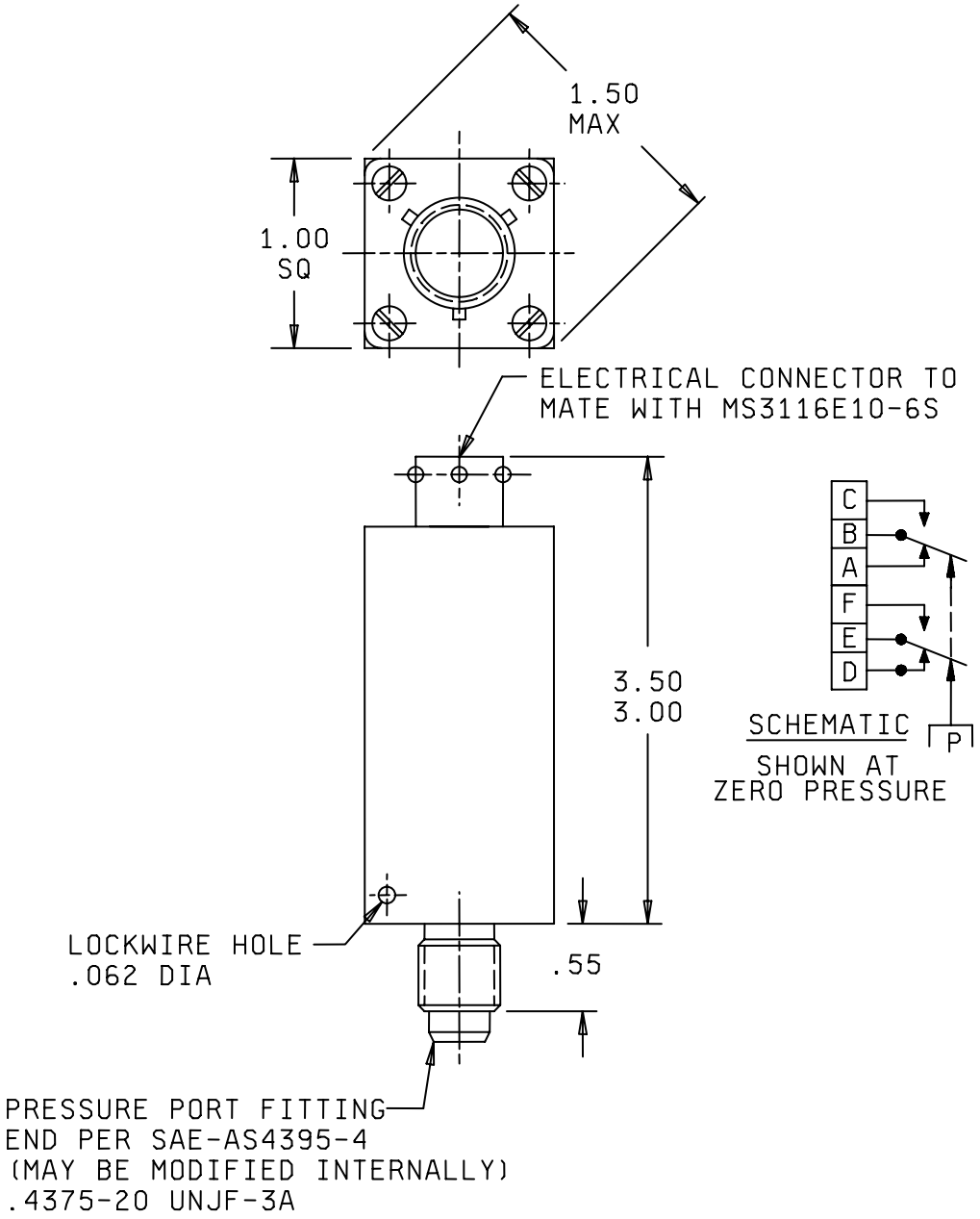


WEIGHT: NOT TO EXCEED .45 lbs.

Configuration 8

FIGURE 1. Switches - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:16Z
Check the source to verify that this is the current version before use.

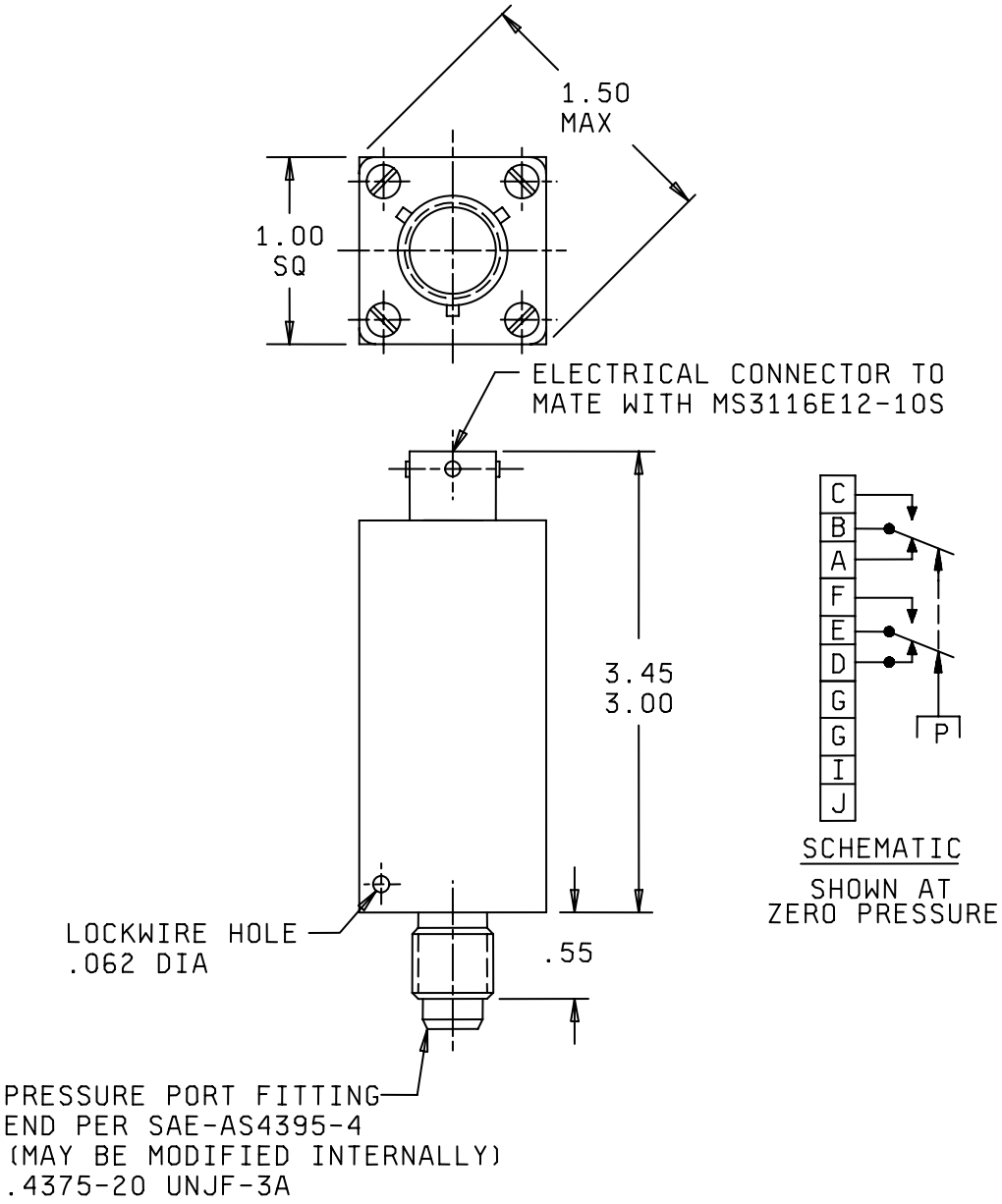


WEIGHT: NOT TO EXCEED .20 lbs.

Configuration 9

FIGURE 1. Switches - Continued.

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WEIGHT: NOT TO EXCEED .20 lbs.

Configuration 10

FIGURE 1. Switches - Continued.

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INCHES	MM
.005	.13
.050	1.27
.062	1.57
.12	3.0
.450	11.43
.55	14.0
.688	17.48
.750	19.05
1.00	25.4
1.38	35.1
1.50	38.1
1.740	44.20
3.00	76.2
3.45	87.6
3.50	88.9

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. Exact shape of switch is optional provided outline dimensions specified are not exceeded and mounting holes and connector locations are as specified.
5. A minimum of two wrench flats is needed.

FIGURE 1. Switches - Continued.

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REQUIREMENTS:

Dimensions, weight, and electrical schematic: See figure 1.

Calibration: See tables I, II, III, and IV.

Proof pressure: 4500 lb_f/in^2.

System pressure: 3000 lb_f/in^2.

Burst pressure: 7500 lb_f/in^2.

Electrical ratings:
Operating voltage: - 28 Vdc.
Current rating - 5 amperes inductive (L/R = .026)
2 amperes lamp load.

Seal:

Pressure chamber: Media proof. Subject switches to proof pressure for 2 minutes using hydraulic fluid per MIL-PRF-5606 with chamber pressure continuously being monitored. Isolate the chamber at proof pressure, with the chamber disconnected from the pressure source. Under that condition, the pressure shall not drop more than 5 lb_f/in^2 for the first 30 seconds to allow for stabilization of test equipment. No pressure loss is allowed thereafter for the remainder of the 2 minutes.

Electrical chamber: Watertight. In the event of primary seal figure, hydraulic fluid shall not enter the electrical chamber.

Electrical connector: See figure 1.

Pressure port: See figure 1.

Media: Hydraulic fluid per MIL-PRF-5606.

High temperature (operating and nonoperating): B (+275°F).

Low temperature (operating and nonoperating): D (-65°F).

Altitude: C (except 80,000 feet).

Shock: C (100 G).

Vibration: S (test condition C, method 204 of MIL-STD-202).

Life (mechanical): F (50,000 cycles).

Life (electrical): B (25,000 cycles).

Acceleration: C (8 G).

Pulsation amplitude: E (10 percent).

Pulsation frequency: D (500 ±50 Hz).

Pressure rise: F (500,000 lb_f/in^2/sec).

Dielectric withstanding voltage (at reduced barometric pressure): Applicable at 250 Vrms.

Connector torque: Applicable.

QUALIFICATION:

Single submission: Restricted to switch submitted.
Group submission: See table V.

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PART NUMBER: Consists of the prefix "M9395/30-" followed by a five-character code. The code identifies the configuration and pressure setting mode (code from table I); high-pressure setting to within 100 lb_f/in^2 (code from table II) followed by high-pressure setting to within 25 lb_f/in^2 and applicable tolerance (code from table III). The five-character code used in the following example identifies a switch of configuration 1 which actuates on increasing pressure at 300 ±25 lb_f/in^2, and deactuates on decreasing pressure at 150±25 lb_f/in^2.

EXAMPLE:

M9395/30- A CA AC

PREFIX

CONFIGURATION AND PRESSURE SETTING MODE= Code "A" of table I.

HIGH-PRESSURE SETTING = Code "C" of table II (300 lb_f/in^2) and Code "A" of table III (0, ±25 lb_f/in^2).

LOW-PRESSURE SETTING = Code "A" of table II (100 lb_f/in^2) and Code "C" of table III (+50, ±25 lb_f/in^2).

TABLE I. Codes for combinations of configurations and pressure setting modes.

	Configuration										Pressure setting mode	
	1	2	3	4	5	6	7	8	9	10	high setting	Low setting
Code	A	D	G	K	N	R	U	X	1	4	At (or max) 1/	At (or min) 1/
Code	B	E	H	L	P	S	V	Y	2	5	At (or max) 1/	Differential 2/
Code	C	F	J	M	Q	T	W	Z	3	5	Differential 2/	At (or min) 1/

1/ Setting values are designated by codes from tables II and III.
2/ Setting values are designated by codes from table IV.

TABLE I. Codes for combinations of configurations and pressure setting modes.

Code	Pressure (lb_f/in^2)	Code	Pressure (lb_f/in^2)	Code	Pressure (lb_f/in^2)
A	100	L	1100	W	2100
B	200	M	1200	X	2200
C	300	N	1300	Y	2300
D	400	P	1400	Z	2400
E	500	Q	1500	1	2500
F	600	R	1600	2	2600
G	700	S	1700	3	2700
H	800	T	1800	4	2800
J	900	U	1900	5	2900
K	1000	V	2000	6-	3000

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TABLE III. Codes for pressure settings to within 25 lb_f/in^2 and tolerances.

	Unit				Tolerance (lb_f/in^2)
	0	+25	+50	+75	
Code	A	B	C	D	±25
Code	E	F	G	H	±50
Code	J	K	L	M	±75
Code	N	P	Q	R	±100
Code	S	T	U	V	±150
Code	W	X	Y	Z	±200
Code	1	2	3	4	±300
Code	5	6	7	8	±400
Code	9	0	I	-	Min or Max

TABLE IV. Codes for differential pressure settings.

Code	Differential value (lb_f/in^2)	Code	Differential value (lb_f/in^2)
A	0	M	275
B	25	N	300
C	50	P	325
D	75	Q	350
E	100	R	375
F	125	S	400
G	150	T	425
H	175	U	450
J	200	V	475
K	225	W	500
L	250		

TABLE V. Extent of qualification .

Part number	Number of samples required	Tests	Qualifies
M9395/30-AADAA ↓ -KADAA -14ZXV -44ZXV	1 each inductive 1 each lamp 1 each inductive 1 each lamp	Complete per qualification inspection of MIL-DTL-9395	ALL configurations

NOTES:

1. Pressure switches supplied to this specification sheet shall be designed to operate normally with the system pump ripple at 3000 lb_f/in^2 at 0 to 700 Hz and average transient of 3000 ±500 lb_f/in^2 at 10 Hz peaking to 4050 lf_f/in^2.
2. Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this spec sheet before specifying a particular "M" number. The fact that operating characteristics can be coded does not necessarily mean that it can be manufactured or procured.

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The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Reference Documents:

Mil-PRF-5606	MS3116	SAE-AS33514
MIL-S-7742	MS3456	
MIL-DTL-9395	SAE-AS4395	
MIL-STD-202	SAE-AS5202	

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-1885)

Review activities:
Army - AV
Navy - MC, SH
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/.

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